MILITARY SPECIFICATION

RECEIVER - TRANSMITTER, RADIO RT-793(*)/ASQ

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1.0 SCOPE

1.1 Scope - The equipment covered by this specification shall provide two-way radio communication in AN/ASQ-19A and AN/ASQ-88A centrals with properly equipped surface stations and other aircraft by means of amplitude modulated radio telephony. When installed with the necessary associated equipment the radio receiver-transmitter shall provide the following:

(1) Communications by means of a-m telephony (A3).

(2) Reception of special-type signals consistent with the performance requirements outlined herein.

(3) Automatic direction finding when installed with certain equipment.

(4) Automatic relay wherein reception of signals on one equipment will cause transmission on another as outlined herein.

1.2 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment:

Antenna AT-14A/A

Headset H-1/AR or H-4/AR

Microphone T-17, NAF-213264-6 or AN8-M-C1 carbon type

Control, Central C-6684/ASQ

Antenna AS-909/ARA-48

Amplifier-Power Supply-Receiver AM-2349/ASQ-19

Intercommunications Station LS-460/AIC

Intercommunications Station LS-459/AIC

Indicator, Frequency Channel ID-1311/ASQ

FSC 5821

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2.0 APPLICABLE DOCUMENTS

2.1 General - The documents listed below of the issue in effect on the date of invitation of bids, form a part of this specification to the extent specified herein:

Applicable Documents

Specifications

Military

MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-I-18309	Integrated Electronic Central AN/ASQ-19A
MIL-I-81036	Integrated Electronic Central AN/ASQ-88A

2.2 Availability of Documents - When requested specifications, standards, drawings and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3.0 REQUIREMENTS

3.1 Preproduction - Preproduction test requirements shall be in accordance with AN/ASQ-19A Specification MIL-I-18309.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing parts and materials shall be in accordance with AN/ASQ-19A Specification MIL-I-18309.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-I-18309 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 38 pounds.

3.3.2 Reliability

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 200 hours without the necessity for readjustment of controls which are inaccessible during flight.

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3.3.2.2 Operating Life - The equipment shall have total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failures (MTBF) - The equipment shall have a specified mean (operating) time between failures of 200 hours when tested and accepted as outlined in AN/ASQ-19A Specification MIL-I-18309.

3.3.3 Cables and Connectors - Connectors used for input wiring or interconnection of removable subassemblies shall be type DA manufactured by the Cannon Electric Company or an approved equivalent. External connectors shall be Amphenal type 126-10572 or an approved equivalent. The wires and terminals of this connector shall be potted with a sealing compound which is capable of withstanding temperatures to 125°C.

3.3.4 Interference Control - The generation of radio interference by the radio set and the vulnerability of the radio set to radio interference shall be controlled within the limits of Specification MIL-I-6181 and MIL-I-18309 with the following exceptions:

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3.3.4.1 Antenna Conducted Keyup Condition - Spurious output between 200 and 500 mc shall not exceed 100 microvolts average and 400 microvolts maximum.

3.3.4.2 Antenna Conducted Key Down Condition - The magnitude of all emitted signals from 200 to 500 mc, inclusive, on other than the carrier frequency shall be 40 db or more below the carrier signal. The magnitude of all emitted signals below 200 mc shall be 70 db or more below the carrier signal. The magnitude of all emitted signals above 500 mc shall be 80 db or more below the carrier signal.

3.3.4.3 Receiver Intermodulation - The magnitude of the receiver intermodulation test signals shall be at least 80 db above one microvolt at the test sample terminal. Receiver sensitivity shall not be degraded by more than 2 db.

3.3.4.4 Susceptibility and Power Line Conducted Interference - Frequency limits shall be 150 kc to 100 mc when testing the equipment. Receiver sensitivity degradation in this range shall be no greater than 2 db. For power line conducted interference, the limits at 25 mc shall apply over the 25 mc to 100 mc range.

3.3.4.5 Radiated Interference - An increase of 16 db above the level stipulated in Specification MIL-I-6181D will be allowed in the frequency range from 80 to 1350 mc. The carrier output frequency may exceed this 16 db limitation by an additional 30 db.

3.3.4.6 Susceptibility, Audio Frequency Conducted - The normal sensitivity of the receiver, which is 4.0 microvolts for radio sets with 4000 cps bandwidth and 4.2 microvolts for radio sets with 6000 cps bandwidth, shall be used as a reference level in performance of this requirement.

3.3.4.7 Images and Other Spurious Responses - When receiving on any channel, the receiver shall attenuate images and other spurious responses to the maximum practicable extent and in any case not less than 70 db relative to the desired signal: except that responses within 500 kc of the desired channel shall not be less than 45 db and shall not be less than 55 db from 500 kc to 1.00 mc relative to the desired signal. For the guard receiver, spurious requirements for frequencies greater than 500 kc from the desired channel shall be governed by MIL-I-6181D.

3.3.4.8 IF Rejection - Under either standard or service conditions, the receiver shall present an attenuation of not less than 80 db, relative to the response of the desired signal to signals on the frequencies of the intermediate frequency amplifier.

3.3.4.9 Cross-Modulation and Desensitization - The cross-modulation and desensitization effects of interfering signals (modulated 30% at 1000 cps) on channels more than 500 kc from and at a level 60 db above the desired channel reference level (4 microvolts) shall permit a signal plus noise-to-noise ratio (desired signal) of at least 6 db. For interfering signals on channels within 500 kc of the desired channel the level shall be 30 db.

3.3.4.10 Short Duration-Interference - Short duration interference limits shall be 20 db above the standard limits of MIL-I-6181 for interference resulting from the manual operation of control switches, external to the RT-793/ASQ, and Spectrum Crystal oven switching. The amount of interference generated by switching the RT-793/ASQ from standby to receive shall have no limitation.

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3.3.5 Service Conditions - The equipment shall operate satisfactorily under any environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-I-18309 for Class I equipment, except as modified herein:

3.3.5.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400. Satisfactory performance when normally mounted (with vibration isolators in place) shall be demonstrated by compliance with the service conditions requirements as specified herein. Satisfactory performance when mounted rigidly (without vibration isolators in place) shall be demonstrated by lack of mechanical failures and satisfactory operation under service condition limits after completion of the vibration tests.

3.3.5.2 Acoustic Noise - No malfunction of the equipment shall occur when subjected to a jet engine exhaust noise of 120 db relative to 0.002 dyne per square centimeter at frequencies between 20 and 10,000 cps.

3.3.6 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-I-18309. The power required shall not exceed the specified amounts:

(1) AC Power (Three Phase) 115 v, 270 VA

(2) DC Power 28 V 7 Amps (Including Channeling)

3.4 Performance

3.4.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply under standard conditions to both main and guard channel operation. Reduced performance and service requirements shall apply only where specifically outlined. Equipment performance under service conditions shall consist of a duty cycle wherein the equipment is operated on a continuous cycle of 10 minutes reception and 5 minutes transmission.

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3.4.2 Frequency Channels - By means of control box selection, it shall be possible to select any frequency in the range of 225.00 - 399.95 mc in 0.05 mc increments. These main channel frequencies shall correspond with the arithmetic series 225.00, 225.05, 225.10, 225.15, ..., 399.95 mc. Additionally, it shall be possible to select a fixed-frequency guard channel in the range of 238.00 mc to 248.00 mc. Unless otherwise specified the equipment shall be supplied with the guard frequency set for 243.00 mc.

3.4.3 Sensitivity - The sensitivity of the receiver shall be such that when an rf input of 4 microvolts, 5 microvolts for guard receiver, (open-circuit) modulated 30% at 1000 cps is applied, the audio output power shall be at least 50 milliwatts with a signal plus noise-to-noise ratio (modulated to unmodulated) of 10 db or greater. When the main receiver is tuned to frequencies within ±5 mc of the guard channel frequency, the main receiver and guard receiver shall meet the sensitivity requirements with 8 microvolts input. (Under service conditions the sensitivity of the main receiver shall not be worse than 7 microvolts, except within ±5 mc of the guard receiver frequency the sensitivity shall not be worse than 10 microvolts. Under service conditions the sensitivity of the guard receiver shall not be worse than 10 microvolts. Under special service conditions of 22 volts dc and 102 volts ac power, the signal plus noise-to-noise ratio shall be at least 10 db for 1000-microvolt signal modulated 30% at 1000 cps.

3.4.4 AVC Characteristics - The AVC action shall be such that when a 1000 microvolt signal modulated 30% at 1000 cps is applied, the audio output power shall be 250 milliwatts (+20%, 59 -10%), and when the signal input (30% modulated at 1000 cps) is varied from 10 microvolts to 100,000 microvolts, the audio output power shall not vary by more than ±3 db from the audio output obtained from the 1000-microvolt signal. When a main frequency within 5 mc of the guard frequency is selected, the foregoing requirement need be met only over the range 20 to 100,000 microvolts. Additionally, the receiver shall be protected against blocking under conditions of input signals up to one-half volt. (Blocking shall be defined as a reduction in audio output to a level below 50 milliwatts) The equipment shall be provided with an internal volume control which shall be adjusted so as to obtain the required output with the 1000-microvolt signal, and then locked before performing other receiver tests.

3.4.5 Selectivity - The over-all selectivity of the receivers shall fall within the limits set forth below for either standard or service conditions.

Attenuation Below Resonant Response	Total Bandwidth Guard Receiver	Main Receiver
6 db	47 kc min.	47 kc min.
60 db	200 kc max.	85 kc max.

3.4.6 Regeneration - The interstage and over-all regeneration of the receiver shall be so minimized that no evidence of unnecessary oscillation, instability, or excessive regeneration shall occur.

3.4.7 Noise Peak Limiter - The detector of the receiver (following the intermediate frequency amplifier) shall be provided with an instantaneous noise peak limiter. The noise limiting action shall effectively limit the audio output voltage of modulating surges or peaks exceeding 50% ±15% modulation of the rf signal. Its action shall be rapid enough to prevent loss of intelligibility due to reduction in receiver gain under the influence of continuous rf noises such as are caused by commutators, ignition systems, etc. When a pulsed signal having a pulse duration of .1 microsecond, a pulse repetition frequency of approximately 500 pulses per second, and peak amplitudes up to 0.1 volt is applied on the same frequency as a desired signal modulated 30% at approximately 1000 cps, not more than 50 microvolts input (desired signal) shall be required to obtain a signal plus noise-to-noise ratio of 6 db.

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3.4.8 Audio Characteristics - The audio characteristics of the receiver shall be as set forth below. Except where otherwise stated, these requirements apply under both standard and service conditions, and apply to signals detected by either the fixed frequency or variable frequency receivers and amplified in the common headset circuit.

3.4.8.1 Capability - Under standard conditions, the receiver shall be capable of delivering an output of at least 250 milliwatts when a signal having any level between 10 and 10,000 microvolts modulated 30% at 1000 cps is applied. With a 1000-microvolt input signal modulated 30% at 1000 cps, the total harmonic content of the receiver output shall not exceed 10% at the 250 milliwatt level. Under service conditions the output shall not be more than 5 db below that obtained under standard conditions. Under the special service conditions wherein the supply voltage is 22 volts d-c and 102 volts a-c, an output of at least 35 milliwatts shall be obtained when a 10,000-microvolt signal modulated 30% at 1000 cps is applied.

3.4.8.2 Over-all Fidelity - The audio response shall not vary more than +1 to -3 db from 300 cps to 4000 cps relative to the response at 1000 cps. Modulation frequencies outside of the 300 to 4000 cps range shall be attenuated the maximum practicable extent without recourse to inclusion in the receiver of large and/or heavy filter components.

3.4.8.3 Auxiliary Circuit - The auxiliary circuit shall deliver only signals detected by the variable frequency receiver. When a 1000-microvolt signal modulated 30% at 1000 cps is applied, the output shall be at least 0.25 volt for a 20,000 ohm resistive load, and when the modulation frequency is varied from 70 to 7000 cps, the output for any freqyency in the range shall be within ±3 db of the output obtained at 1000 cps.

3.4.9 Carrier-to-Noise Ratio Squelch Characteristics - A carrier-to-noise ratio squelch circuit shall be incorporated in both the main channel and guard channel receivers. Internal squelch threshold level controls shall be provided in both receivers. Under standard conditions the adjustment range of each control shall be such that the squelch threshold can be set to correspond to an output signal plus noise-to-noise ratio between the limits of 3 db and 10 db when the r-f signal input of less than 4.0 microvolts is modulated 30% at 1000 cycles per second. The squelch circuit shall have positive action so that the receiver audio is either completely off or completely on at all times. Unless specified otherwise the squelch shall be adjusted to open at a 6 db signal plus noise-to-noise ratio, ± 1 db. Under service conditions with no rf signal input the squelch shall not open.

3.4.10 RF Power Output - The transmitter shall be capable of delivering the following values of radio frequency carrier energy within the range described below:

Standard Conditions	NLT 20 watts average (16 watts minimum)
Service Conditions	7 watt min.
Reduced Performance Test at 102 v a-c and/or 22 v d-c.	3 watt min.

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3.4.11 Modulated Characteristics - The transmitter shall meet the following requirements for modulation when operated under both standard and service conditions.

3.4.11.1 Modulation (Radio Telephony A-3) - Under standard conditions the modulator shall be capable of modulating the r-f carrier output of the transmitter not less than 80% when the input voltage at 1000 cps obtained from the dummy microphone (per Paragraph 3.5.1.10.4) is adjusted to 1 volt (open circuit) for the carbon microphone and to 0.15 volt (open circuit) for the dynamic microphone.

Modulation shall be determined from oscilloscope presentation of the modulated carrier wave and from measurement of maximum and minimum voltage displayed on the oscilloscope. Modulation shall be calculated from the formula:

$$\frac{E_{max}-E_{min}}{E_{max}+E_{min}} \times 100\%$$

The modulator of this equipment shall employ clipper or limiter provisions for clipping positive and negative peaks at a modulation level of not more than 95% with increased microphone input voltage.

The modulator gain control shall be adjusted to provide modulation to the clipping level with 1 volt (open circuit) input for the carbon microphone and 0.15 volt (open circuit) input for the dynamic microphone. With the input reduced 3 db below the clipping level, the distortion of the 1000 cps modulation envelope shall not exceed 10%. At this level the transmitter shall have the following fidelity characteristics relative to the output at 1000 cps.

Frequency, cps	Relative output in db
100	-5 or lower
300	0 to -4
600	+1 to -3
1,000	0 (ref. level)
3,000	+1 to -3
4,000	+1 to -3

3.4.11.2 Modulation Under Service Conditions - Under service conditions the modulator shall be capable of modulating the r-f carrier output of the transmitter at least 70%. Distortion of the modulation envelope with the input 3 db below the clipping level shall not exceed 10%.

3.4.11.3 Microphone Current - The carbon microphone energizing current measured with dummy microphone connected and the equipment operated from a 27.5 volt d-c supply shall be 50 $\pm$7 milliamperes.

3.4.11.4 Tone Modulation - Not less than 80% modulation of the r-f carrier shall occur when tone transmission is selected and the transmitter is Keyed. The frequency of modulation shall be between 800 and 1200 cps. For service conditions the percentage modulation shall be not less than 70%.

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3.4.12 Frequency Stability - The frequency stability of the complete equipment after operation under either standard or service conditions after the required warm-up time shall be such as to maintain the equipment frequencies within the limits shown below provided the ambient temperature does not exceed 55°C.

Transmitter carrier frequency tolerance	±10 kc
Fixed - frequency receiver resonant frequency tolerance	±13 kc
Variable - frequency receiver resonant frequency tolerance	±10 kc

3.4.13 Sidetone - Monitoring of the carrier modulation shall be effected via the receiver audio output channel. The signals delivered to the receiver output while the transmission is in effect shall be gated by rectifying a small portion of the carrier output. Under standard conditions the minimum sidetone output from the receiver shall be approximately 250 milliwatts when the transmitter carrier is modulated 80% at 1000 cycles. Under service conditions the output shall not be less than 50 milliwatts. The total harmonic content of the output shall not exceed 15%, with the sidetone level 3 db below clipping. A sidetone gate shall be provided which will open the sidetone circuit when the rf output falls to a level between 0.1 to 0.5 watt, to indicate to the operator the loss of adequate rf power output.

3.4.14 Internally Generated Noise - Under standard conditions, and with no modulation voltage applied transmitter carrier noise shall be at least 40 db below the maximum modulation obtainable. The receiver signal plus noise-to-noise ratio shall be at least 30 db when a 1000 microvolt signal modulated 30% at 1000 cps is applied and the total receiver output (signal plus noise) is adjusted to 250 milliwatts. Under service conditions, the receiver signal plus noise-to-noise ratio shall be at least 10 db when a 10 microvolt signal modulated 30% at 1000 cps, is applied.

3.4.15 Channel Changing Cycle - The time elapsing between the instant the channel selector switch is positioned to select a given channel and the instant the equipment is in an operative condition on the selected channel shall not be greater than 6 seconds under either standard or service conditions. This requirement shall hold for shifting the operating frequency of the equipment between any 2 channels. During channel changing operations, the output of the receiver (headset circuit) shall indicate that the equipment is not in readiness for transmission should the operator begin speaking before the channel changing cycle is complete. During channel change a 400 cps audio tone shall be applied to the audio output.

3.4.16 Frequency Modulation - Under the service conditions and with no modulation voltage applied to the transmitter and no rf signal applied to the receiver the peak frequency deviation arising from frequency modulation of the receiver conversion oscillator or the transmitter carrier shall not exceed 1.0 kc. Under the standard conditions, and with the emitted carrier modulated 80% at 1000 cycles, the peak frequency modulation of the transmitter shall not exceed 1.0 kc.

3.4.17 Transmit-Receive Interval - When shifting from transmission to reception or vice versa on the same channel (with an rf signal input voltage of 1000 microvolts modulated 30%), not more than 100 milliseconds shall elapse between the operation of the control switch and the time the equipment is stabilized in the selected mode of operation in compliance with the performance requirements. This requirement shall be met under both standard and service conditions.

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3.4.18 Accuracy of Automatic Tuning - When the same frequency channel is selected several times in succession, the transmitter r-f power output and the receiver audio power (with an input signal of 100 microvolts or greater modulated 30% at 1000 cycles) shall not vary by more than 0.5 db from the power outputs obtained initially. This requirement shall be met under standard or service conditions provided such conditions are static.

3.4.19 Relaying - When two equipments are used in an automatic relaying installation, the reception of a RF signal on one receiver-transmitter shall cause the other receiver-transmitter to emit an RF signal at some preselected frequency, other than the received frequency. The detailed performance requirements shall be as follows:

3.4.19.1 Single-Channel Communication - When transmission and reception only are selected on a single set (either receiver-transmitter), performance shall be as required by this specification for one receiver-transmitter in a nonrelaying installation.

3.4.19.2 Dual Reception, Single Transmission - When reception on both sets and transmission on one set is selected, the performance rendered by the sets shall be as required by this specification for one set operating singly in the mode of operation selected, (transmission or reception) except that the receiver audio power output (or sidetone) may decrease not more than 2 db.

3.4.19.3 Relaying Operation - The application to the antenna circuit of either radio set of a signal on the receiver resonant frequency and of a level sufficient to fully open the receiver squelch circuit shall cause the other radio set to generate and emit a radio-frequency carrier wave of the rated power on the selected frequency, and removal of the actuating signal shall cause cessation of the generated carrier. When the actuating signal is 3 db or more above the minimum level at which relaying in accordance with the foregoing sentence is possible, the generated carrier shall reach the rated power level within 150 milliseconds after application of the actuating signal and the system shall be returned to standby condition within 150 milliseconds after removal of the actuating signal.

3.4.19.4 Relaying Modulation - When the equipment is relaying in accordance with the requirements of the foregoing paragraphs and amplitude modulation is applied to the actuating signal, the emitted carrier shall be modulated in an essentially similar manner. When a signal 3 db above the level required to actuate the squelch (with squelch adjusted to open at a 6 db signal plus noise-to-noise ratio) is applied and is modulated 80% at 1000 cycles, the emitted carrier shall be modulated to the extent specified in 3.4.11.

3.4.19.5 Relaying Monitor - When the equipment is relaying as described above, the operator headsets of both equipments shall be connected so as to monitor all signals relayed in either direction. The operator of either equipment can disable the automatic relaying operation and transmit by pressing push-to-talk switch. Under these conditions, the audio power output may decrease not more than 6 db. Distortion and signal plus noise-to-noise ratio at the terminals of the headset circuit, and r-f carrier output and modulation shall be essentially the same as those obtained when ordinary operation is selected.

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3.4.20 Automatic Direction Finding - When installed with certain associated equipment, the radio-set shall be suitable for automatic direction finding. When used for this service, the receiver-transmitter will receive radio frequency signals from the direction finding antenna, amplify and demodulate them and deliver the low-frequency modulation components to the direction finder amplifying and indicating equipment. When the radio set is connected in an automatic relaying installation, the automatic direction finding function is available on one receiver-transmitter. The auxiliary audio output (see 3.4.8.3) shall be in phase +0.28 milliseconds with the negative portion of the rf signal envelope of a 1000 microvolt signal modulated with a 200 cps square wave.

3.4.21 RF Power and VSWR Indicator - A meter shall be provided to determine the approximate RF carrier power output with the RT-793/ASQ in the transmit (keyed) mode. By pressing an indicator button the reflected rf power can be read from this meter. This meter shall be calibrated in steps from 0 to 25 watts.

With the antenna terminal of the RT-793/ASQ terminated with a 50 ohm non reactive load, the power meter shall be accurate to within +5.0 watts between 225.00 mc and 399.95 mc; and the reflected power indicated by the meter shall be not more than 3.0 watts.

3.4.22 Auxiliary Power - The equipment shall contain an auxiliary dc power source used to energize the C-6684/ASQ control box. The power source shall be capable of supplying a voltage of not less than +29 volts nor more than +35 volts, and shall be capable of providing up to 17.5 watts maximum.

3.5 Detailed Requirements

3.5.1 Receiver-Transmitter, Radio RT-793/ASQ - The receiver-transmitter shall meet the following requirements.

3.5.1.1 Function - The receiver portion shall receive rf energy from the antenna, amplify and convert it to aural intellignece. The transmitter portion shall generate rf power and transmit it to the load specified herein.

3.5.1.2 Form Factor - The receiver-transmitter shall be contained in a single unit capable of being directly mounted to a shockmounted equipment shelf which in turn is mounted in an aircraft. It shall be possible to dismount and remove it from its case using ordinary hand tools. The overall dimensions of the receiver-transmitter shall not exceed 7.403 inches in height, 11.545 inches in width, and 14.165 inches in length.

3.5.1.3 Warning Notice and Mechanical Clearance - The following notice shall be placed on the plenum assembly: "Mounting surface of modules must be flush with chassis before tightening mounting screws. Adequate clearance shall be provided to allow for ease of adjustment of all circuits during electrical alignment procedures."

3.5.1.4 Contents - This unit shall contain the complete transmitter, modulator, direct crystal-controlled frequency generator, automatic tuning devices, frequency selection system, RF power and VSWR indicator, and all other parts to make the equipment function as a unit for the auxiliary equipment as specified in paragraph 1.2. It shall be possible to remove or replace any plug-in unit located on the main chassis using ordinary hand tools. It shall be further possible to replace any plug-in unit with another without resynechronization except that realignment of tuning elements may be necessary. The above circuits shall be contained in the following eleven (11) modular assemblies:

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(1) Power Amplifier - Provides RF amplification during transmission; and during receive condition, is used as the first mixer stage.

(2) Spectrum Generator - Provides high frequency injection.

(3) Mechanical Tuner - Tunes RF, oscillator, first and second IF circuitry by positioning the gears connecting these modules.

(4) Power Supply - Converts three-phase-115VAC power to D.C. to provide the control (C-6684/ASQ), transmitter and receiver plate and bias voltages.

(5) First & Second IF - Consists of two tunable IF amplifier sections used for transmission and reception.

(6) Third IF - Provides third IF amplification, carrier-to-noise squelch system, and detector circuitry.

(7) Modulator & Audio Amplifier - Amplifies microphone audio and modulates RF carrier of transmitter, and provides receiver audio amplification for the variable frequency & guard receivers.

(8) Guard Receiver - Provides emergency receiver circuitry except for audio amplification.

(9) Receiver Preamplifier - Amplifies RF input signals during reception and provides preamplification of RF signals during transmission.

(10) Static Frequency Converter - Converts binary Frequency Control-box (C-6684/ASQ) codes to the two-out of five (2/5) code compatible with the equipments tuning system.

(11) RF Power and VSWR Indicator - Provides an indication of the amount of RF (average) power transmitted and reflected power due to antenna mismatch during transmitter operation.

3.5.1.5 Electrical Connection

3.5.1.5.1 External

Preference Designation	Receptical Type	Function
J 11	Amphenal 126-10572 (or approved equivalent)	Power Signal Control RF Connection

3.5.1.5.2 Internal - Internal Connectors per paragraph 3.3.3.

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3.5.1.6 Controls, Internal - The following semi-fixed adjustable controls shall be accessible with the unit removed from its case.

(1) Main receiver sensitivity control, 1A3R31

(2) Guard receiver sensitivity control, 1A7R46

(3) Main audio amplifier gain control, 1A4R21

(4) Guard audio amplifier gain control, 1A4R20

(5) Modulator audio gain control, 1A4R1

(6) Side tone audio gain control, 1A4R42

(7) Squelch control, 1A3R50, 1A7R51

3.5.1.7 Finishes, External - Equipment covered by this specification shall be painted with color number 26231 of Federal Standard 595, dated 1958.

3.5.1.8 Pressurization - Items of equipment required by this specification to be sealed shall comply with Specification MIL-E-5400 except that a desiccant is not required, a safety valve is not required, mechanical pressurization pumps shall not be used in the equipment and the cases shall be pumped with dry air to a differential of 2 to 4 pounds per square inch at sea level pressures and room ambient temperatures. Loss of pressure shall not exceed 5 pounds per square inch for a 24 hour period at 50,000 feet altitude and +55°C ambient temperature.

3.5.1.8.1 Air Pressure Indicator - A non-air release type pressure indicator shall be mounted in the front of the cover of the Receiver-Transmitter and shall give a visual indication when the internal air pressure is adequate for proper operation of the equipment.

3.5.1.9 Cable Length - Under either standard or service conditions, the equipment shall operate as required by this specification when using control cables up to 25 feet in length fabricated from #22 AWG copper wire, or when using cables up to 110 feet in length fabricated from #16 AWG copper wire.

3.5.1.10 Terminations

3.5.1.10.1 Receiver RF - RF input to the receiver shall be through a 50-ohm cable from a signal generator having an effective source impedance of 50-ohms. Sensitivity shall be in terms of "open-circuit" microvolts.

3.5.1.10.2 Transmitter RF - The RF output load (dummy antenna) shall consist of a 50-ohm nonreactive load which shall include means suitable for measuring the RF Power output, percentage modulation and distortion.

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3.5.1.10.3 Receiver Audio - The main (headset) audio output channel of the receiver shall be terminated in a 300 ohm nonreactive load which shall include means for measuring the audio-frequency power output. The auxiliary (wide-range) audio output channel for the receiver shall be terminated in a resistance load of 20,000 ohms, including means for measuring the audio output voltage.

3.5.1.10.4 Transmitter Input - The audio (microphone) input circuit of the transmitter shall be connected to an audio-frequency generator and dummy microphone having characteristics as listed below:

Carbon Microphone Input

Resistance - 82 ohms
Impedance (mainly resistive at audio frequencies) - 82 ohms
Frequency range - 150 - 10,000 cps
Total harmonic distortion - less than 1%
Open-circuit output potential (adjustable 0.05 to 3.0 Vrms)

Dynamic Mike Input

Resistance - 82 ohms
Impedance (mainly resistive at audio frequencies) - 82 ohms
Frequency range - 150 - 10,000 cps
Total harmonic distortion - less than 1%

The audio input circuit shall provide full performance capability for 0.150 Vrms open circuited output potential. The audio input circuit to the input transformer shall be a balanced input type and shall be open circuited during the receive mode.

4.0 Quality Assurance Provisions - Equipments shall be subjected to the tests of specification MIL-I-18309 to determine compliance with all the applicable requirements.

5.0 Preparation for Delivery

5.1 General - Preparation for delivery shall be in accordance with specification MIL-I-18309.

6.0 Notes

6.1 Use - This equipment is intended to supply the UHF communication function for the integrated Electronic Centrals AN/ASQ-19A or AN/ASQ-88A.

6.2 Precedence - When the requirements of the contract, this specification or applicable subsidary specifications are in conflict, the following procedure shall apply.

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(1) Contract - The contract shall have precedence over any specification.

(2) System Specification - Integrated Electronic Central AN/ASQ-19A Specification MIL-I-18309 shall have precedence over this specification.

(3) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the procuring activity.

(4) Reference Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 v dc and 115 +1.0 v ac as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptance standard of performance.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other then those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.6 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with AN/ASQ-19A Specification MIL-I-18309. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number and date of this specification.
(2) Selection of applicable levels of packaging and packing (see 5.1).

*U.S. GOVERNMENT PRINTING OFFICE: 1967-253-223/769

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-R-81443(AS) RECEIVER-TRANSMITTER, RADIO RT-793 (*)/ASQ

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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